

6(a)	bronchi / left and right bronchus ;	1
6(b)	<i>any two from:</i> <u>diffusion</u> distance too great / AW ; A walls too thick for (efficient) <u>diffusion</u> <i>ref. to</i> surface area time taken for <u>diffusion</u> (to occur) too long / <u>diffusion</u> rate too slow ; further detail ; e.g. many cell layers / a number of tissues, to pass across named tissue type(s) to cross e.g. presence of smooth muscle <i>ref. to</i> ciliated epithelium / not squamous epithelium <i>ref. to</i> layer of mucus on lining of gas exchange system <i>idea that</i>, ventilation / fast air flow, reduces time for (correct gradient to be established for) diffusion to occur ; <i>idea that</i> blood supply is, not from pulmonary artery / not pulmonary circulation / from a branch of the aorta / part of the systemic circulation ; I not enough, blood / blood vessels detail ; e.g. (systemic) blood supply to the gas exchange organs is, already oxygenated / not deoxygenated low / no, concentration gradient for, oxygen / carbon dioxide in (systemic) blood supply to the gas exchange organs	2
6(c)(i)	as airway generation (number), increases / gets higher, airway diameter decreases ; AW A in terms of named structures or numbers A inverse correlation	1

Question	Answer	Marks
6(c)(ii)	<i>any three from:</i> <i>yes, useful</i> compared to Weibel Lung model measurements, similar results / same trend / same relationship between airway generation and airway diameter ; data to support ; projection / multi-Slice, for at least one generation, compared to Weibel (supporting trend) projection / multi-slice, 2 generations showing trend no risk of, harmful / radiation, exposure ; AW detail ; e.g. can repeat / can use regularly (on same person) can, follow progression of disease <i>if no reference made to 'yes' results, only allow one of these two marks*</i> <i>not useful</i> not the same as / different to, Weibel's results ; projection results not consistent with multi-slice results / AW ; data to support ; <i>not same as Weibel's</i> projection or multi-Slice, for at least one generation compared to Weibel <i>projection and multi-slice not the same</i> projection and multi-Slice for at least one generation	3